

# FRAME: Consumer-Conditioned Memory Serving for World Models and Physical AI

An audited formulation, layout-serving study, and learned-policy validation program

Pengfei Xiao

Project page: [suspectpfx.github.io/frame-world-model-memory](https://suspectpfx.github.io/frame-world-model-memory) · September 2026

**Abstract.** Memory for embodied agents is often summarized by bytes, reconstruction error, or a single retrieval score. These quantities do not determine whether a record can be interpreted by a particular policy, world model, planner, or verifier at the time it is used. We formulate **FRAME**, a consumer-conditioned memory-serving problem that separates representation fidelity, semantic or spatial granularity, physical layout, transform metadata, record age, and temporal acquisition channel. A record is admissible only when it can be constructed from resident information or an explicitly permitted sensor acquisition; a serving decision is then selected on the Pareto frontier of resident bytes, latency, and energy under consumer loss and deadline constraints.

We audit the mechanisms required by this formulation. On a frozen spatial reader, a hidden reversible permutation increases absolute latent drift from 0.022 to 0.440, while retaining and applying the inverse restores 0.022 and reproduces the natural arm exactly in the stored scores. The reported values 1.073 and 1.121 are traced to ratio-of-means and mean-of-ratios estimands. A CPU/GPU A–G layout study spans 76 configurations per device and finds implementation-dependent operating points; 0 of 760 exploratory contrasts survive a conservative Holm family. Finally, 480 synchronous LIBERO episodes with a learned Cosmos Policy deployment show that stale inputs change executed actions and task outcomes, and inverse-permutation access is action-equivalent to the matching TTL control across 120 paired states. These runs validate integration and measurement but do not contain a calibrated FRAME selector or policy-consumed learned imagination. The closed-loop Pareto-improvement hypothesis remains open.

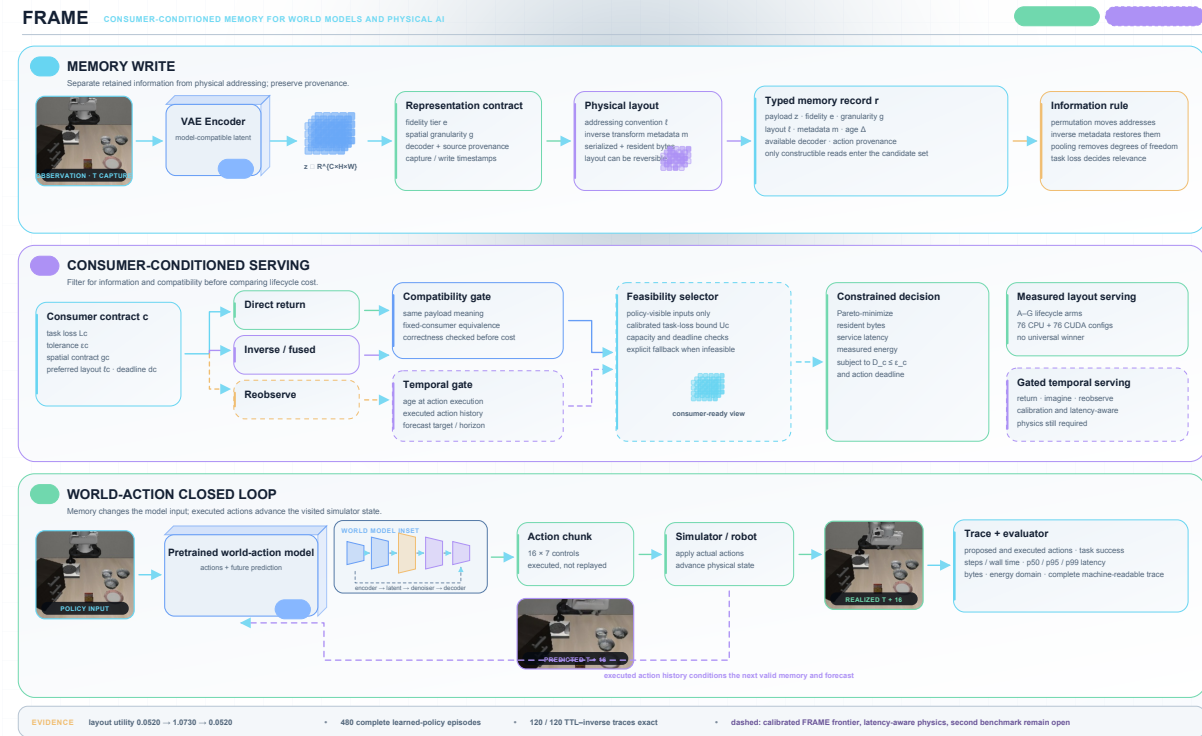


Figure 1: FRAME writes typed records, serves named consumers, and closes the world-action loop; dashed modules remain gated experiments.

**Evidence boundary**

The manuscript separates algebraic implications, retrospective exploratory results, and future confirmatory hypotheses. Supported means regenerated from source artifacts within the stated consumer, deployment, and sample unit. It does not imply suite-wide or hardware-independent generality.

## 1 Introduction

World models and vision-language-action policies use memory through concrete input contracts. A spatial reader may require the correspondence between a feature vector and its grid cell. A verifier may tolerate spatial aggregation yet fail under stale object state. A policy may accept an old image, a learned prediction, or a newly sensed observation, but those alternatives have different information, latency, and energy consequences. A memory system that reports only compression ratio or reconstruction error therefore omits the variable that decides utility: the downstream consumer.

FRAME treats memory as a serving problem. The system records what information is retained, how it is addressed, when it was captured, how it can be adapted, and which temporal channel produced the input. A consumer supplies a task loss, input schema, fidelity or spatial contract, preferred physical layout, and deadline. The runtime searches only feasible constructions and returns a Pareto-minimal alternative under measured resource constraints. This formulation makes three distinctions explicit.

First, a reversible layout transformation and a lossy spatial transformation have different failure modes. A permutation with missing metadata causes *addressability failure*: information remains present, but the reader cannot associate values with their intended locations. Pooling or downsampling can cause *information loss*: multiple inputs share one stored representation, and task-relevant distinctions may be unrecoverable. Second, the capture time of a record remains attached to the information even after decoding, transformation, or prediction. Third, a memory decision becomes an embodied-systems claim only when it changes the model input, the executed action, and the visited state in a closed loop.

The paper makes four contributions, with their evidential status stated directly:

1. We give a consumer-conditioned formulation that separates representation tier, granularity, physical layout, transform metadata, age, access adapter, and temporal channel. This is a research hypothesis whose task-level value must be tested against fixed policies and conventional caches.
2. We establish, for the audited frozen reader, the predicted signature of addressability failure and exact metadata-aware recovery. We also show that the reader-dependent granularity knee is distinct from a conservative high-fidelity margin. **Supported within scope.**
3. We implement seven layout-serving arms on CPU and GPU and measure correctness, storage, write/read latency, tails, throughput, and board/package energy where available. The observed operating points depend on fusion and hardware; a universal break-even is not supported. **Supported within scope.**
4. We integrate a learned action-conditioned Cosmos Policy variant into synchronous LIBERO, execute paired memory controls, and record task, latency, energy, and action traces. This validates the experimental chain while leaving calibrated selection, learned imagination consumed by the policy, latency-aware physics, and cross-engine replication unfinished. **Open.**

## 2 Related Work and Boundary of the Contribution

**Memory in action-conditioned world models.** Mem-World retrieves historical observations through a wrist-view-centered surfel memory to improve persistent future prediction [7]. Echo-Memory controls backbone and training factors while comparing capacity, compression, read-out, and recurrence for return consistency [4]. These works ask which history improves world prediction and how memory mechanisms affect generation. FRAME addresses a neighboring systems question: given heterogeneous consumers and resource constraints, in what representation, granularity, layout, age, and access path should a record be served? The contribution does not rely on the broad claim that world models previously lacked memory.

**Embodied benchmarks and latency semantics.** LIBERO provides manipulation suites designed to study knowledge transfer [5]. RoboTwin 2.0 provides domain-randomized bimanual tasks and a scalable generator [1]. ReflexBench decouples simulator stepping from control and exposes configurable synchronous and asynchronous latency [2]. FRAME uses LIBERO for the current integration and adopts latency-aware evaluation as a required validation protocol; it does not claim to introduce those semantics. RoboTwin replication remains a planned cross-engine test.

**Physical layout planning.** Compiler layout selection jointly accounts for operator preferences and explicit conversions in a dataflow graph [3]. FRAME adds semantic task utility, spatial information granularity, temporal age, and sensor or imagination

channels to the memory-serving decision. The classical layout optimization problem remains a subproblem rather than a new FRAME theorem.

**Action-conditioned model integration.** The current learned-policy deployment follows the Cosmos Policy action/future interface [6]. Because the official tokenizer artifact was access-restricted in the available environment, the experiments use a separately published Wan VAE compatibility variant. Its identity with NVIDIA’s tokenizer is unestablished, and every result is labeled accordingly.

### 3 Problem Formulation

#### 3.1 Memory record and provenance

For sensor field  $j$ , FRAME stores

$$\mathbf{r}_j = (z_j, e_j, g_j, \ell_j, m_j, t_j^{\text{cap}}, t_j^{\text{write}}, p_j), \quad (1)$$

where  $z_j$  is the payload;  $e_j$  is the representation or fidelity tier;  $g_j$  is semantic or spatial granularity;  $\ell_j$  is the physical addressing convention;  $m_j$  contains shape, decoder, and inverse-transform metadata; and  $p_j$  records provenance. Capture and write time are distinct. Re-encoding or reindexing changes the write time but cannot make sensed information younger.

A write operator has the generic form

$$z_j = P_{\ell_j}(Q_{e_j, g_j}(E(o_j))), \quad (2)$$

where  $E$  is a model-compatible encoder,  $Q$  controls retained representation and granularity, and  $P$  chooses physical layout. Varying  $e$  and  $g$  jointly does not identify their separate effects; the evaluation varies them independently whenever the consumer interface permits it.

Predicted records additionally retain the originating observations, model identifier, prediction target time, and conditioning action history. A future conditioned on a proposed action sequence is reusable only when its conditioning history agrees with the actions that were actually executed.

#### 3.2 Consumer contract and access

A consumer  $c$  specifies

$$C_c = (\mathcal{L}_c, \epsilon_c, s_c, g_c, \ell_c, d_c), \quad (3)$$

where  $\mathcal{L}_c$  is the task loss,  $\epsilon_c$  a tolerance,  $s_c$  the input schema,  $g_c$  a spatial or semantic contract,  $\ell_c$  a preferred physical layout, and  $d_c$  an action deadline. A preferred layout is a performance preference: an exact adapter can satisfy the input contract without preserving that layout at rest.

For policy  $\pi$  at time  $t$ , the accessed representation is

$$\hat{z}_{c,t} = \text{Access}_{\pi}(\mathbf{r}_j; c, t) = T_{\ell_j \rightarrow \ell_c}(z_j; m_j), \quad (4)$$

with adapter  $a \in \{\text{direct}, \text{materialize}, \text{fused-indirect}\}$ . Materialization constructs a contiguous canonical tensor. A fused adapter permits indexed loads inside the consumer. Correctness for the fused path concerns consumer output,

$$\text{Consume}_{\text{fused}}(z, m) \doteq \text{Consume}(\text{Recover}_{\ell, m}(z)), \quad (5)$$

within a declared numerical tolerance and with identical reduction semantics.

#### 3.3 Temporal channels and use-time age

FRAME separates three temporal sources:

$$k \in \{\text{return}, \text{imagine}, \text{reobserve}\}. \quad (6)$$

Return adapts a resident observation or valid prediction. Imagine advances resident information with a learned action-conditioned model. Reobserve opens a sensor gate, acquires specified fields, timestamps them, encodes them, and may write a new record. The policy cannot receive simulator ground truth unless reobservation is explicitly selected.

The age actually seen by the consumer is

$$\Delta_{j,t}^{\text{use}} = t^{\text{action}} - t_j^{\text{cap}}, \quad (7)$$

where  $t^{\text{action}}$  includes queueing, access, model inference, and sensing delay under latency-aware evaluation.

### 3.4 Consumer distortion and constrained choice

For deployment distribution  $Q$  and a fixed consumer, define

$$\mathcal{D}_c(\pi; Q) = \mathbb{E}_{(I_t, Y_c) \sim Q} [\mathcal{L}_c(Y_c, f_c(\text{Access}_\pi(I_t)))]. \quad (8)$$

The paper states whether  $\mathcal{L}_c$  is an absolute prediction loss, an excess loss relative to fresh input, or an episodic task loss. These estimands cannot be interchanged.

FRAME selects a Pareto-minimal feasible policy:

$$\begin{aligned} & \underset{\pi \in \Pi_{\text{constructible}}}{\text{Pareto-minimize}} && \left( B_{\text{resident}}(\pi), T_{\text{service}}^{95}(\pi), E_{\text{episode}}(\pi) \right), \\ & \text{s.t.} && \mathcal{D}_c(\pi; Q) \leq \epsilon_c, \\ & && B_{\text{resident}}(\pi, t) \leq B_{\text{max}} \quad \forall t, \\ & && \Pr[T_{\text{action}}(\pi) > d_c] \leq \alpha_c. \end{aligned} \quad (9)$$

A runtime uses a calibrated upper bound  $U_c(q \mid I_t)$  based only on policy-visible information; the evaluator measures actual violations. If no candidate is feasible, a predeclared fallback is invoked and logged. Arbitrary weighted sums are unnecessary unless the deployment supplies weights with operational meaning.

## 4 Three Mechanistic Implications

### 4.1 Reversible addressability

Let  $P$  be a bijective permutation and let  $m$  identify its inverse. Then

$$P_m^{-1} P z = z. \quad (10)$$

An identical deterministic consumer consequently returns the same output. If  $m$  is missing, a spatial reader may fail because values are associated with the wrong locations. A permutation-invariant reader may remain unchanged. The algebra is established mathematics; the scientific question is whether metadata and access cost matter in the measured system.

### 4.2 Lossy granularity

A many-to-one coarsening  $Q_g$  maps at least two inputs to one payload. General input recovery is therefore impossible. Task information is lost only when  $Q_g(x_1) = Q_g(x_2)$  while the consumer requires different outputs for  $x_1$  and  $x_2$ . Consequently, lossy representation and insufficient for consumer  $c$  are different statements.

Two empirical quantities must also remain separate. The *hard collapse threshold* is the point where the store no longer resolves distinctions required by the reader. A *conservative high-fidelity margin* is an operating region chosen to keep loss below a practical bound. The current data suggest a reader-relative knee and a post-hoc finer-grid margin; they do not establish a universal 2× rule.

### 4.3 Conditional reuse crossing

For reuse count  $F$ , compare canonicalization on write with inverse reindexing on each read:

$$C_{\text{preserve}}(F) = C_{\text{canonicalize-write}} + F C_{\text{direct-read}}, \quad (11)$$

$$C_{\text{reindex}}(F) = C_{\text{raw-write}} + F(C_{\text{metadata}} + C_{\text{inverse}} + C_{\text{consumer}}). \quad (12)$$

Under a fixed affine model, the difference is  $\delta_{\text{write}} + F \delta_{\text{read}}$ . A positive crossing exists only when slopes and intercept support it and the crossing lies in the measured range. Cache state, fusion, batching, and queuing can violate this model. Failure to resolve a crossing is a valid result.

## 5 Runtime and Experimental Design

### 5.1 Closed-loop causal chain

At each decision, FRAME reads record timestamps and the consumer contract; enumerates constructible representation, access, and temporal choices; screens them with a calibrated, policy-visible feasibility model; and selects a Pareto-minimal candidate.

The action/world model receives that input, proposes an action chunk and optionally a future, and the simulator executes the selected controls. The actual action history conditions the next decision. The evaluator alone receives ground truth.

The complete trace records requested and served schemas, payload and metadata bytes, capture and action times, adapter identity, proposed and executed actions, simulator transition, success, latency components, energy domain, model and source hashes, and fallback events. This trace is the unit that connects the memory decision to a physical outcome.

## 5.2 Hypotheses and statistical protocol

Four hypotheses are declared:

### H1 — consumer utility.

Minimal feasible representation and age regions differ among named consumers.

### H2 — addressability.

An unknown reversible layout degrades a spatial reader, while metadata-aware recovery meets a predeclared practical equivalence margin.

### H3 — serving trade-off.

Preserve-layout and reindex-on-read occupy different storage, latency, and energy operating points in a frozen deployment family.

### H4 — embodied frontier.

A calibrated FRAME policy improves the closed-loop task/resource Pareto frontier under a learned action-conditioned model.

The present analyses are retrospective and exploratory. Future confirmatory runs freeze endpoints, practical margins, task selection, and code before collecting results. Paired initial conditions are used when possible. Task is the higher-level unit for claims across tasks; episodes are nested within task. Success uses hierarchical bootstrap or mixed-effects logistic regression, with Holm step-down within each confirmatory family. Effect sizes and 95% intervals accompany  $p$ -values. Non-significance is never treated as equivalence.

## 5.3 No-current-state-information controls

A frame sampled from the same sequence can retain scene identity, background, geometry, and task context. It is therefore labeled a within-sequence time-shuffled baseline. A cross-sequence prior supplies a stronger no-current-state-information control. Both appear in the reader study so that layout effects are not interpreted against a mislabeled zero-information baseline.

## 5.4 Layout-serving arms

The A–G study evaluates: (A) canonical layout with direct read; (B) hidden permutation with discarded metadata as a diagnostic negative control; (C) permutation plus metadata with materialization on read; (D) permutation plus fused indirect gather; (E) canonicalization on write; (F) a shared canonical store with per-consumer adapters; and (G) replicated consumer-specific layouts. Before profiling, every non-diagnostic arm verifies reconstruction and fixed-consumer output correctness.

The sweep varies payload, channel rank, spatial cells, reuse  $F \in \{1, 2, 4, 8, 16, 32, 64, 128\}$ , consumers, working-set size, cache treatment, and CPU versus GPU. The primary latency summary is the median across randomized run blocks, with paired bootstrap intervals. Energy blocks remain separated by measurement domain.

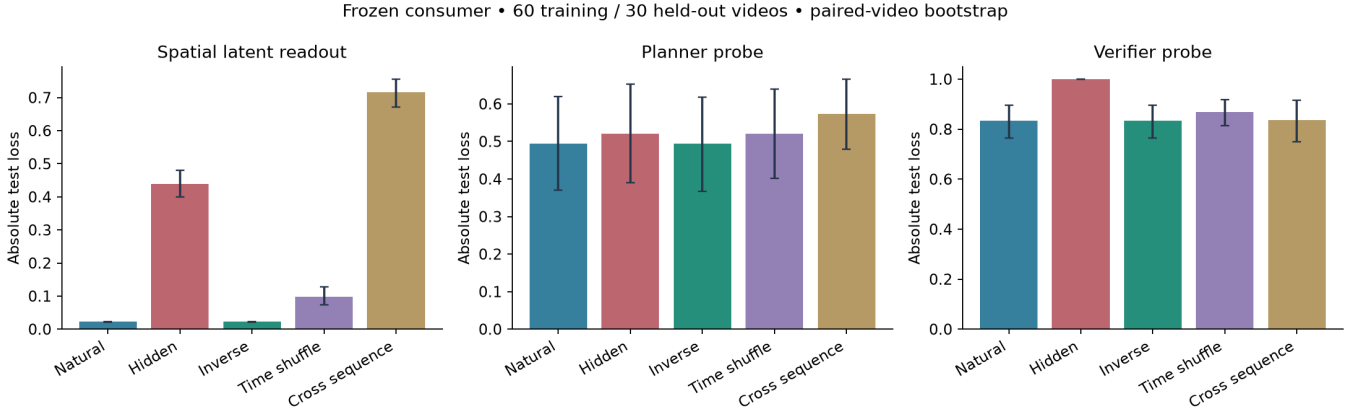
# 6 Addressability and Granularity Results

## 6.1 Hidden layout mismatch and exact recovery

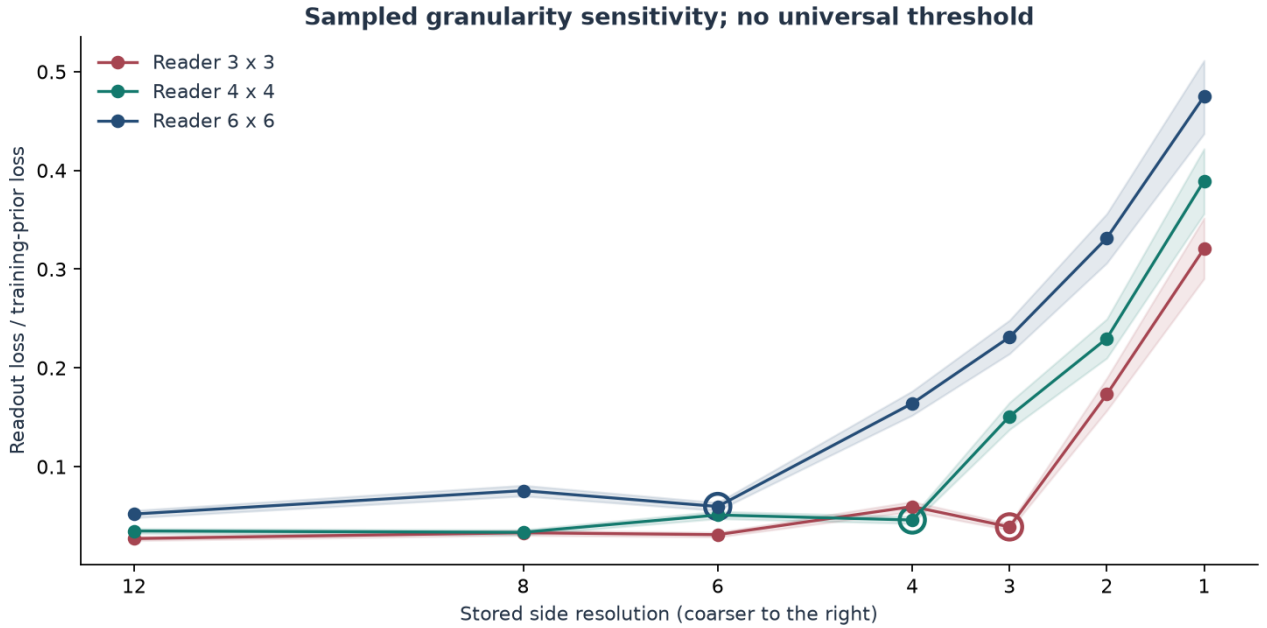
The frozen-reader study uses 60 training and 30 held-out videos. The spatial latent reader is trained once on natural layout, then evaluated without refitting across natural, hidden-permutation, inverse-recovered, time-shuffled, and cross-sequence conditions. As shown in Fig. 2, absolute latent drift rises from 0.022 [0.021, 0.024] under natural layout to 0.440 [0.399, 0.480] when the permutation is hidden. Applying the retained inverse returns 0.022 [0.021, 0.024]. The stored per-video inverse scores equal the natural scores with maximum absolute discrepancy 0.

The project page previously displayed normalized hidden-layout errors near 1.073 and 1.121. Regeneration resolves the discrepancy: 1.073 is the ratio of mean loss to mean prior loss, whereas 1.121 is the mean of within-video ratios. Natural and inverse-recovered values are 0.052 and 0.052 under the former estimand, and 0.053 and 0.053 under the latter. The numbers answer different questions and must carry their estimand labels.

For a 144-cell permutation, 104 bytes is the fixed-length coding lower bound, not the measured implementation footprint. The serialized index artifact occupies 144 bytes and the resident int64 inverse map occupies 1152 bytes. Historical average inverse



**Figure 2:** Frozen-reader response to reversible layout mismatch and two no-current-state-information controls. Error bars are paired-video bootstrap 95% intervals. The spatial reader is sensitive to addressability; the planner and verifier probes are much less diagnostic in this experiment.



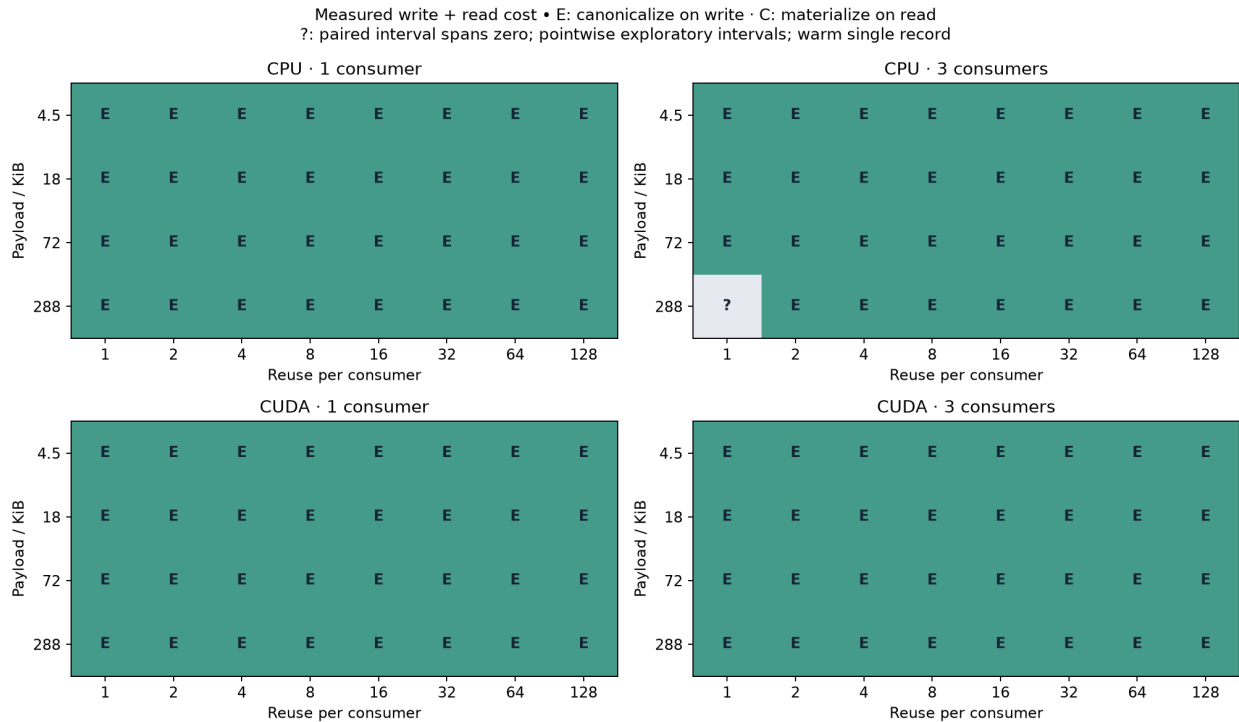
**Figure 3:** Store granularity versus frozen reader resolution. Each curve has its own knee; shaded intervals are paired-video bootstrap intervals. The experiment separates a reader-relative collapse threshold from a finer operating margin.

timing near 9.7–10.0  $\mu\text{s}$  is retained as a bounded microbenchmark. The associated modeled 0.0017 mJ value is not an independently integrated operation-level energy measurement and is excluded from the main energy claim.

## 6.2 The collapse knee follows the reader

Figure 3 varies store grid while holding reader grids at  $3 \times 3$ ,  $4 \times 4$ , and  $6 \times 6$ . Error rises when the stored grid falls below the reader’s own resolution, and the knee shifts with the reader. This supports a consumer-relative spatial contract within the probe family. It does not identify a universal threshold, and the post-hoc region at roughly twice the reader resolution is reported only as a candidate conservative margin.

The historical claim  $p = 0.019$  cannot be audited because the original null, tail, seed, and sample construction are missing. A later cell-level reconstruction gives nearby values but violates independence because cells share videos. The result is therefore removed rather than reinterpreted. Retrospective video-level tests use a declared four-test family and Holm correction; they remain exploratory because hypotheses and thresholds were selected after inspecting the data.



**Figure 4:** Exploratory phase map for canonicalization on write (E) versus materialization on read (C). A question mark denotes a paired interval spanning zero. The grid is a measured deployment slice, not a universal law.

## 7 Preserve Layout Versus Reindex on Read

The study contains 76 CPU and 76 GPU configurations, 5852 randomized run blocks per device, and 456 correctness checks per device. CPU and GPU implementations are each internally paired. Cross-device differences combine hardware and implementation effects because the GPU version fuses indirect access for both D and F, while the CPU version uses tiled or materialized operators in some arms.

Figure 4 compares canonicalization on write (E) with materialization on read (C). E has lower pointwise measured write-plus-read cost across nearly the entire warm single-record grid, but those colored cells are exploratory paired intervals. The more consequential result appears in Fig. 5: on CPU, materialization often beats indirect gather, while on GPU the fused indirect path often wins. Shared canonical storage versus replicated layouts is less stable and contains unresolved cells.

Across all 760 server contrasts, 0 exact paired sign-flip tests survive Holm correction at 0.05. With eleven run blocks, the exact test has coarse resolution, and counting pointwise colored cells does not produce a confirmatory claim. The evidence supports three conclusions: metadata-aware access is correct; materialized and fused paths are materially different implementations; and no universal break-even or universally negligible reindex cost has been established. Advisor feedback on storing the inverse and comparing lifecycle cost is therefore closed for the audited deployment scope, while generalization to larger records, independently randomized per-record mappings, queuing, and further hardware remains open.

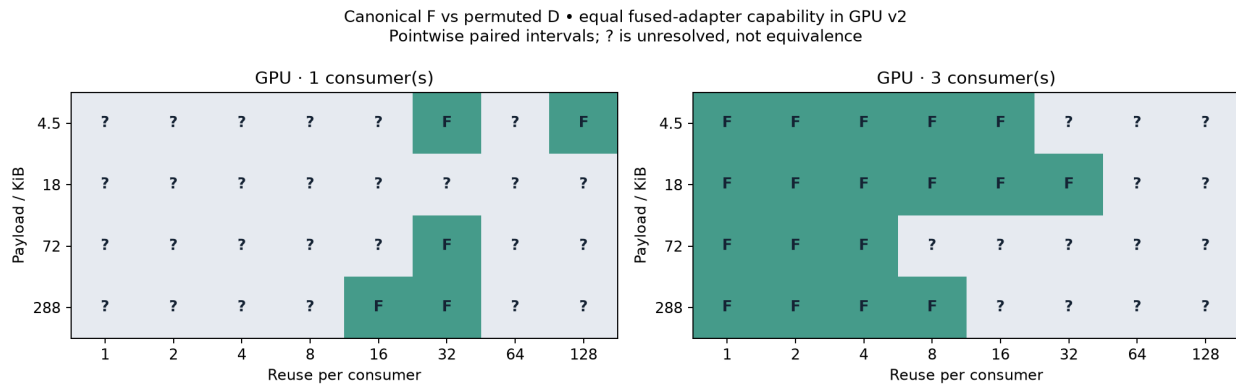
## 8 Learned-Policy LIBERO Evidence

### 8.1 Deployment and episode population

The integration pins Cosmos Policy source revision 18a2accadf4e7a3531e56754102af5a24d2316da and loads the published action-model weights on an NVIDIA RTX A6000. An inference-only compatibility port replaces unsupported attention kernels without changing checkpoint keys or shapes. A public Wan VAE supplies the tokenizer/decoder path; its SHA-256 is retained, and equivalence to the access-restricted official tokenizer is not claimed.

The learned model emits finite  $16 \times 7$  action chunks, predicted future images, and value estimates. In the population study, the policy consumes RGB and proprioception inputs and executes actions in LIBERO. Learned future images are logged as evaluator outputs; they are not yet selected as policy inputs. Simulator stepping is synchronous, so physics is frozen during inference. These conditions define the present evidence boundary.

The completed table contains 480 episodes: three fixed tasks, 40 paired initial states per task, and four arms. Fresh acquires a



**Figure 5:** Fused and materialized access comparisons. Equal optimization changes the apparent winner, especially on GPU. Physical layout therefore remains an explicit serving cost; the data do not justify treating it as a mandatory stored property.

**Table 1:** Learned-policy LIBERO controls. Success intervals are exact binomial 95% intervals within each fixed task; wall time and energy are medians. Energy is gross allocated-GPU-board energy and excludes sensing, CPU, and robot energy.

| Task    | Arm           | Success | Acq. | Wall (s) | GPU (kJ) |
|---------|---------------|---------|------|----------|----------|
| Spatial | Fresh         | 40/40   | 5.0  | 45.3     | 5.35     |
| Spatial | Stale         | 0/40    | 1.0  | 82.3     | 18.73    |
| Spatial | TTL           | 2/40    | 7.0  | 86.7     | 19.48    |
| Spatial | TTL + inverse | 2/40    | 7.0  | 86.7     | 19.52    |
| Object  | Fresh         | 40/40   | 9.0  | 49.5     | 6.85     |
| Object  | Stale         | 0/40    | 1.0  | 91.8     | 13.20    |
| Object  | TTL           | 12/40   | 9.0  | 97.3     | 13.70    |
| Object  | TTL + inverse | 12/40   | 9.0  | 97.5     | 13.69    |
| Goal    | Fresh         | 40/40   | 8.0  | 56.7     | 7.22     |
| Goal    | Stale         | 0/40    | 1.0  | 134.3    | 18.09    |
| Goal    | TTL           | 40/40   | 5.0  | 67.8     | 9.19     |
| Goal    | TTL + inverse | 40/40   | 5.0  | 67.4     | 9.14     |

current observation at every policy decision; stale retains the first observation; TTL refreshes periodically; and TTL plus inverse stores the same TTL inputs in a permuted layout with retained inverse metadata. All episodes use official simulator success and save action, observation, timing, energy, and provenance traces.

## 8.2 What the controls establish

Fresh observation succeeds in 40/40 states on each fixed task. Stale input succeeds in 0/40. TTL succeeds in 2/40 Spatial, 12/40 Object, and 40/40 Goal states. The paired TTL-minus-fresh success differences are  $-0.95$  with CI  $[-1.00, -0.875]$ ,  $-0.70$  with CI  $[-0.825, -0.55]$ , and  $0.00$  with CI  $[0, 0]$ , respectively. These fixed-task effects show that memory age changes task outcomes and that one refresh schedule is not uniformly adequate.

TTL plus inverse exactly matches TTL across all 120 paired states: input hashes, action shapes, and executed action arrays agree, with maximum absolute action difference zero. This carries inverse recovery through the actual learned policy and simulator action chain. It does not demonstrate a task advantage for inverse access; exact recovery should make the two arms equal.

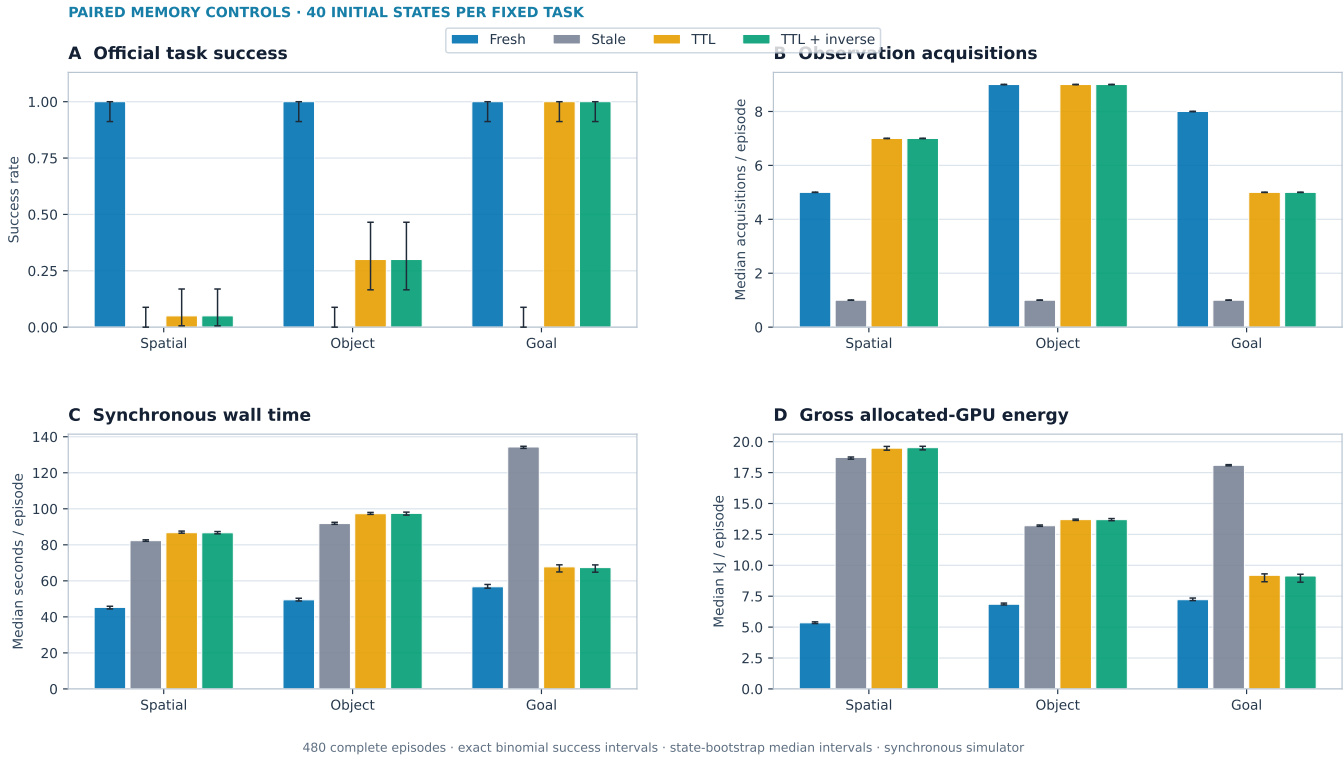
Action disagreement relative to fresh input is high after trajectories diverge, confirming that the intervention changes executed behavior. After divergence, disagreement combines input and visited-state effects and is not a one-step causal estimand. Gross GPU-board energy excludes CPU, sensing hardware, display, and physical-robot energy. Fresh episodes were run earlier than memory controls, so machine-load and run-order effects remain. All energy values are labeled by measurement domain.

## 8.3 World-model probe

Figure 7 displays observation, predicted future, realized future, and error views from three preselected smoke episodes. The probe verifies a real learned action-conditioned future pathway. It does not validate the selector’s ability to decide when imagination is task-sufficient because the policy does not consume those predicted futures.

**Table 2:** Paired TTL effect relative to fresh observation. Intervals resample the 40 fixed initial states within a task; they do not support suite-wide generalization.

| Task    | Difference | 95% paired CI    | Action disagreement |
|---------|------------|------------------|---------------------|
| Spatial | -0.95      | [-1.000, -0.875] | 0.794               |
| Object  | -0.70      | [-0.825, -0.550] | 0.885               |
| Goal    | 0.00       | [0.000, 0.000]   | 0.869               |



**Figure 6:** Paired learned-policy memory controls over 480 synchronous LIBERO episodes. Bars show success, observation acquisitions, wall time, and gross allocated-GPU-board energy. TTL is a fixed control, not the proposed calibrated FRAME selector.

## 9 What Is Established and What Remains Open

### 9.1 Advisor feedback on layout metadata

The inverse-permutation question is answered at three levels. Algebraically, bijective recovery is exact when metadata is retained. Empirically, the frozen spatial reader recovers its natural-layout score. Operationally, the A–G study compares direct, materialized, fused, canonicalize-on-write, shared-store, and replicated-layout paths across reuse and payload regimes. The study finds implementation-dependent costs and no confirmatory universal crossing. The design implication is to retain layout as a costed serving choice; the data do not support enforcing consumer layout for every stored record.

### 9.2 Advisor feedback on world models and Physical AI

The current work uses a real learned action-conditioned model, executes its actions in closed-loop LIBERO, reports official success, end-to-end synchronous wall time, model latency, GPU peak memory, and labeled GPU-board energy, and verifies that memory interventions alter actions. These results satisfy integration, instrumentation, and causal-chain prerequisites.

The main scientific criterion remains unmet: no calibrated FRAME selector chooses among return, imagination, and reobservation on held-out data, and learned predicted futures are not consumed by the policy. Latency also does not advance physical state in the current synchronous simulator, and no second benchmark has been run. H4, a FRAME improvement in the closed-loop task/resource Pareto frontier, therefore remains **Open**. The paper does not replace that missing comparison with presentation material or aggregate microbenchmarks.



**Figure 7:** Learned future-prediction probe. Predicted and realized frames are aligned to executed action chunks; pixel and patch errors are evaluator outputs. The figure validates integration and exposes prediction errors, while policy-consumed imagination remains an open experiment.

### 9.3 Novelty statement

The narrow novelty target is a systems formulation and runtime that jointly condition memory representation, spatial information, physical addressability, temporal age, and acquisition or imagination channel on an explicit downstream contract, then evaluate the resulting choice in an action loop. Each ingredient has precedents. The testable contribution lies in the joint decision boundary and in showing when measured properties simplify it. Current experiments fully support the addressing mechanism and deployment study, and they establish the prerequisites for the embodied test. They do not yet establish that the joint policy expands the task/resource frontier.

## 10 Confirmatory Validation Program

### 10.1 Stage 1: freeze the selector and endpoints

Use disjoint training, calibration, and test trajectories. Fit  $U_c(q \mid I_t)$  only from policy-visible features such as record age, representation identifiers, model-prediction diagnostics available online, and consumer identity. Freeze thresholds, tie-breaking, fallback behavior, and all four confirmatory families before test execution. Report calibration coverage and constraint failures rather than silently relaxing  $\epsilon_c$ .

### 10.2 Stage 2: paired LIBERO pilot

Run one Spatial, one Object, and one Long/Goal task with 30–50 paired seeds per task. Required arms are fresh observation, stale retrieve, fixed TTL, fixed fidelity, FRAME preserve-layout, FRAME reindex-on-read, FRAME without consumer conditioning, FRAME without imagination, a lightweight predictor, and real-model imagination. All arms use the same learned policy, decoder, action horizon, and success semantics. Scaling is permitted only if the baseline reproduces expected behavior, interventions change actions, no ground-truth leakage exists, timing is stable, and success is statistically estimable.

### 10.3 Stage 3: latency-aware execution

Add an asynchronous evaluator in which sensing, adapter, world-model, and policy latency advance simulator time under a predeclared action-hold rule. Report synchronous and latency-aware results separately. The primary embodied endpoint is task success; secondary endpoints include completion, actions or steps to success, wall-clock time, deadline misses, reobservations, imagination steps, age distribution, model error, bytes, latency tails, and measured energy per decision, episode, and successful episode.

### 10.4 Stage 4: replication and systems baselines

After pilot gates pass, expand across LIBERO suites and replicate direction on 5–10 RoboTwin 2.0 tasks covering pick/place, open/close, bimanual coordination, handover, and long-horizon execution. The memory store must also compare LRU, LFU, ARC or 2Q, size-aware eviction, frequency-size heuristics, global fixed fidelity, and per-consumer fixed fidelity under identical capacity and traces. A cross-task claim uses task as the higher-level unit. No-crossing, no-gain, and negative replication outcomes remain reportable.

## 11 Limitations

The frozen-reader experiments use learned probes rather than the end task of a robot policy. Their conclusions are consumer-specific and retrospective. The layout sweep covers arbitrary cell permutations and bounded payloads; it does not include all transpose, block, tiling, cache, or multitenant queueing regimes. CPU and GPU versions differ in fusion details, preventing a pure hardware causal comparison.

The learned-policy evaluation covers three fixed LIBERO tasks and a compatibility tokenizer variant. The fresh arm saturates on those initial states, while the TTL control fails sharply on two tasks. This identifies sensitivity but leaves little room for estimating a positive improvement over fresh sensing. A resource frontier requires a calibrated selector that saves sensing or compute while maintaining task quality. The current energy domain is the allocated GPU board, and synchronous physics omits control delay. RoboTwin, conventional cache baselines, policy-consumed imagination, and hierarchical cross-task inference remain absent.

The claim registry contains 608 rows: 112 confirmed within explicit scope, 459 provisional, 28 not supported, 2 contradicted, and 7 retracted. Registry entries include repeated figure and table checks and bounded arithmetic assertions; their counts are an audit trail, not a count of scientific discoveries.

## 12 Conclusion

FRAME defines memory quality through the consumer that uses a record and the physical loop in which that use occurs. The formulation separates retained information from addressability, access implementation from at-rest layout, and information age from write time. Audited evidence shows that hidden layout metadata can produce a large spatial-reader failure while exact inverse access restores utility, and that the cost of recovery depends on fusion and deployment. Learned-policy LIBERO controls verify that age and layout interventions reach executed actions and simulator outcomes.

The strongest defensible conclusion is conditional: consumer contracts and explicit adapters are necessary to state the memory-serving problem correctly, and their mechanisms have been validated in the measured probes and systems. A calibrated joint FRAME policy has not yet been shown to improve the closed-loop task/resource frontier. Completing that test requires held-out selector calibration, policy-consumed imagination, latency-aware physics, and cross-engine replication.

**Table 3:** Machine-readable claim registry after the audit. The four confirmatory hypotheses remain unrun as confirmatory families.

| Status                        | Registry rows |
|-------------------------------|---------------|
| Confirmed within stated scope | 112           |
| Provisional / exploratory     | 459           |
| Not supported                 | 28            |
| Contradicted                  | 2             |
| Retracted                     | 7             |
| Total                         | 608           |

## A Claim Registry and Evidence Policy

Every headline claim is linked to a dataset, consumer, backbone, protocol, source script, source artifact, sample unit, effect, interval, test family, and status. Final figures are generated from raw or processed tables. A result is retracted when provenance is insufficient or a stronger rerun contradicts it. Negative outcomes are retained in the failure log.

The four H1–H4 entries are confirmatory hypotheses in the registry and remain unrun as confirmatory families. Retrospective analyses can calibrate a future design but cannot be relabeled after inspection. The historical  $p = 0.019$ , 104 B measured-metadata claim, universal 2× requirement, and end-to-end energy-saving claims were revised or removed for this reason.

## B A–G Layout Arm Definitions

**Table 4:** Serving arms used in the layout experiment. B is a diagnostic control and is excluded from performance correctness comparisons.

| Arm | Name                   | Operation                                                                                 |
|-----|------------------------|-------------------------------------------------------------------------------------------|
| A   | Canonical direct       | Store directly in the consumer layout and read without conversion.                        |
| B   | Hidden permutation     | Store a permuted payload and discard the map; diagnostic addressability failure.          |
| C   | Materialize on read    | Retain metadata, gather into a contiguous canonical tensor, then run the consumer.        |
| D   | Fused indirect         | Retain metadata and consume through indexed loads without full canonical materialization. |
| E   | Canonicalize on write  | Convert once when the record enters memory; subsequent reads are direct.                  |
| F   | Shared canonical store | Maintain one shared representation with equally optimized per-consumer adapters.          |
| G   | Replicated layouts     | Store a physical copy for each consumer-specific layout.                                  |

Correctness is checked before profiling. For reversible permutations, the payload invariant is  $\|P_m^{-1}(P_m(z)) - z\|_\infty < \tau$  with  $\tau = 0$  for the integer-indexed test payload and a declared floating-point tolerance for consumer outputs. Accuracy must match the canonical arm before latency and energy measurements are admitted.

## C Reproducibility Record

The project archive records configuration, source hash, environment, job identifier, raw traces, processed tables, plot scripts, and statistical reports. Successful population jobs were 4523–4525; dependency-gated analysis job 4535 combined them with the fresh baseline. Failed renderer launches 4512–4514 exited before model execution and contribute no episodes. The executed population-source SHA-256 is

851041b59cca827112b4a1fce013b5719656a299bb6dbe912759653b005459d5.

The model checkpoint is 3,913,017,345 bytes; the VAE is 507,609,880 bytes. Peak allocated GPU memory is reported from the process allocator, and energy is associated with the CUDA device UUID. The public project page exposes generated tables, figures, videos, registry records, and scope labels. Full per-step server traces remain in the recorded experiment directory because of their size.

## D Required Trace Schema for the Confirmatory Study

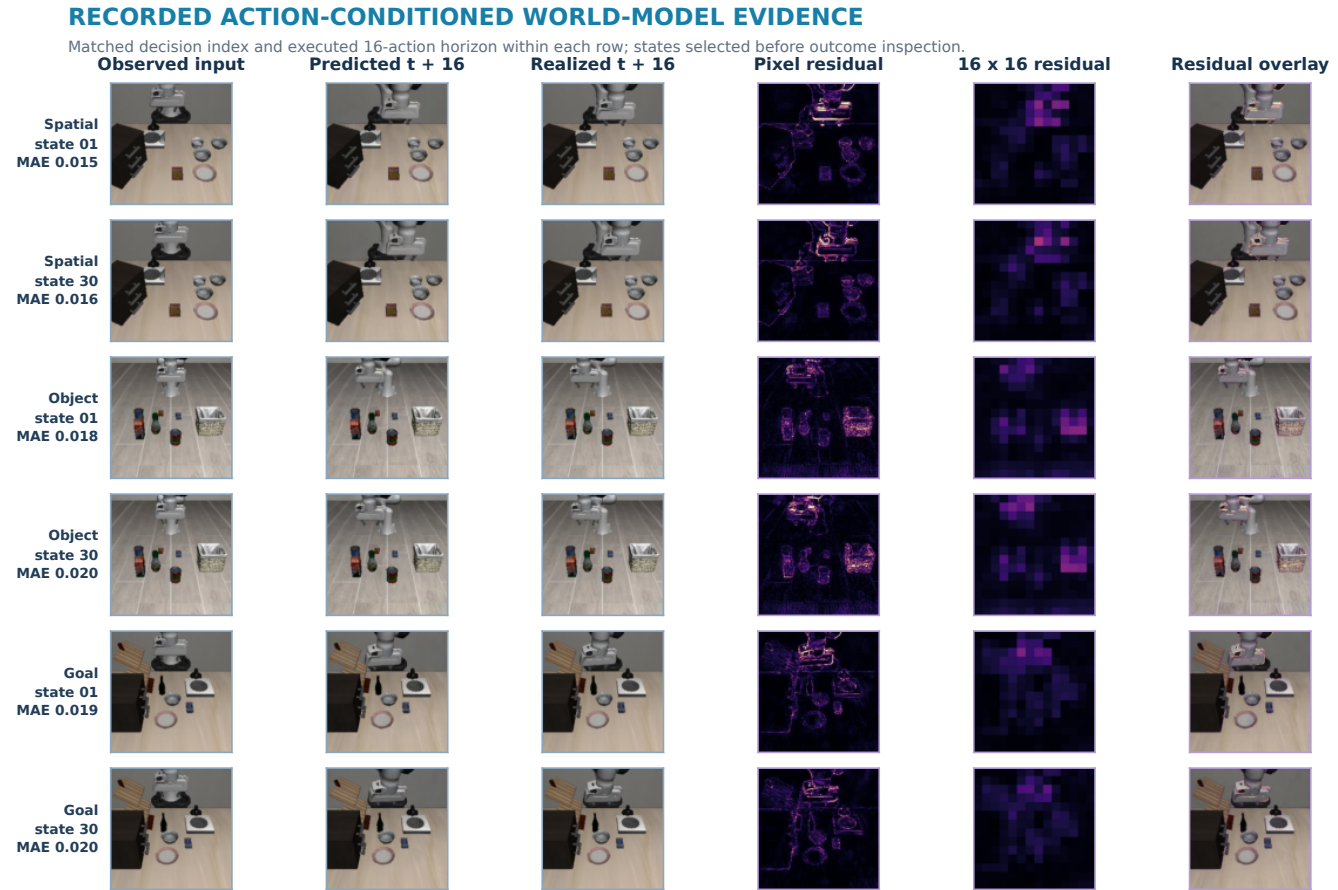
Each decision record must include episode, task, seed, simulator state checksum, consumer identity, request schema, loss tolerance, deadline, record identifiers, representation tier, spatial granularity, layout, metadata bytes, capture time, write time, age at use, adapter, temporal channel, policy-visible selector features, predicted feasibility, selected fallback, proposed action, executed action,

model future, simulator transition, official success, latency components, deadline result, bytes moved, memory occupancy, energy domain, energy sample, model hash, code hash, and configuration hash.

Ground-truth state may be stored for evaluation but is unavailable to the selector and policy unless the declared observation channel exposes it. The trace validator checks this separation, verifies action-conditioning history for imagined records, and rejects runs with missing or mismatched hashes.

## E Full Visual Evidence Matrix

Figure 8 places matched observation, prediction, realized future, and residual views from the recorded action-conditioned model on one page. Each row shares a decision index and the same executed 16-action horizon. The matrix is qualitative evidence of model execution and error structure; it is not a selected success demonstration or a substitute for task-level statistics.



Source: server-generated LIBERO trajectories and learned Cosmos Policy futures · qualitative evidence only

**Figure 8:** Recorded learned-model visual evidence across Spatial, Object, and Goal tasks. Columns show current observation, predicted future, realized future, pixel residual, patch residual, and residual overlay.

## F Main Confirmatory Tables to Produce

The final H4 study should produce four primary tables from raw traces:

1. task-level success and completion with task-clustered intervals;
2. observation, imagination, and adapter use with complete byte accounting;
3. p50, p95, and p99 action latency plus deadline-miss probability under synchronous and latency-aware execution;
4. measured energy per decision, episode, and successful episode, separated into GPU board, CPU package, DRAM, sensing, and total domains where instrumentation permits.

Primary figures should show success versus end-to-end latency and energy Pareto frontiers, with nondominated points computed only after quality constraints are applied. A method that lowers cost by violating its consumer loss tolerance is infeasible rather

than Pareto-superior.

## References

- [1] Tianxing Chen, Zanzin Chen, Baijun Chen, Zijian Cai, Yibin Liu, Zixuan Li, Qiwei Liang, Xianliang Lin, Yiheng Ge, Zhenyu Gu, et al. RoboTwin 2.0: A scalable data generator and benchmark with strong domain randomization for robust bimanual robotic manipulation. *arXiv preprint arXiv:2506.18088*, 2025.
- [2] Yuxuan Chen, Wanruo Zhang, and Xiao Li. Reflex: Enabling fast and predictive vision-language-action models for reaction-critical manipulation. *arXiv preprint arXiv:2608.14379*, 2026.
- [3] Clemens Eisenhofer, Yuwen Jia, Daniel Kroening, and Sergey Pupyrev. Tensor seeks layout: Formalizing layout selection for ML compilers. *arXiv preprint arXiv:2608.21555*, 2026.
- [4] Wayne King, Zeyue Xue, Yuxuan Bian, Jie Huang, Haoran Li, Yaowei Li, Yaofeng Su, Yuming Li, Haoyu Wang, Shiyi Zhang, et al. Echo-Memory: A controlled study of memory in action world models. *arXiv preprint arXiv:2606.09803*, 2026.
- [5] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. *arXiv preprint arXiv:2306.03310*, 2023.
- [6] NVIDIA. Cosmos policy: Fine-tuning video models for visuomotor control and planning. Cosmos Cookbook, 2026.
- [7] Zirui Zheng, Jiaqian Yu, Xiongfeng Peng, Jun Shi, Mingyi Li, Chao Zhang, Weiming Li, Dong Wang, Huchuan Lu, and Xu Jia. Mem-World: Memory-augmented action-conditioned world models for persistent robot manipulation. *arXiv preprint arXiv:2606.18960*, 2026.